One shared block does direct deduction, 0.011 less accurately

Stage P's censored tied cell, re-run at three widths with the sharing held fixed:
d192 -> d320 -> d448, applied 16 times.

S 51853969 | commit d9b177f | 3 seeds/cell | N=48 d5 q4 --shuffle-adj --fall | 14 rho bins x 400 instances | 240k train, 50 epochs

Stage P could not tell a damaged algorithm from a small one

- Tying 16 blocks into 1 at d192 censored: no rho bin cleared 0.5
- A censor is not a number, and two readings fitted it equally
- Only buying capacity back at FIXED sharing separates them

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	`BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING`
	`BEGIN_ROUND SOLVED UNRESOLVED`
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	`COLOR_0 COLOR_1 COLOR_2 COLOR_3`
	the q colours (token name COLOR_n, prose 'colour')
10-57	`N0 N1 ... N47`
	one id per node, no structure in the id

GRAMMAR

`BEGIN_HINTS`	`(NODE COLOR)*`
`BEGIN_GRAPH`	`(NODE (NODE){degree})*`
`BEGIN_REASONING`	`(BEGIN_ROUND (NODE COLOR)+)*`

`SOLVED`

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 393 TOKENS

HINTS	`BEGIN_HINTS N1 COLOR_2 N3 COLOR_1 N4 COLOR_3 N6 COLOR_3 N7 COLOR_3 N8 COLOR_0 ... (59 tokens)`
GRAPH	`BEGIN_GRAPH N0 N47 N3 N26 N33 N43 N1 N23 N32 N11 N12 N38 ... (289 tokens)`
REASONING	`BEGIN_REASONING BEGIN_ROUND N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 N17 COLOR_1 N23 ... (45 tokens)`

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 5 rounds, 19 deductions,
38 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000011, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

29 hints of 48 nodes, then 5 rounds of simultaneous deduction. Hints: N1=COLOR_2 N3=COLOR_1 N4=COLOR_3 N6=COLOR_3 N7=COLOR_3 N8=COLOR_0 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N5 COLOR_3 N11 COLOR_3 N14 COLOR_0 N16 COLOR_2 ...
  11 deduced
BEGIN_ROUND        N18 COLOR_0 N25 COLOR_2 N35 COLOR_1 N47 COLOR_3
  4 deduced
BEGIN_ROUND        N0 COLOR_2 N19 COLOR_0
  2 deduced
BEGIN_ROUND        N46 COLOR_2
  1 deduced
... 1 further round

SOLVED
```

WHY ONE DEDUCTION FIRES

node N5, neighbours N2 N15 N9 N13 N36

their colours known before this round:

```
   [0, 1, 2]  (3 = q-1 of the 4)

   (4 known neighbours, 3 DISTINCT colours -- some repeat)

   still unknown: N2
```

only colour left: 3

```
so the trace emits  N5 COLOR_3
```

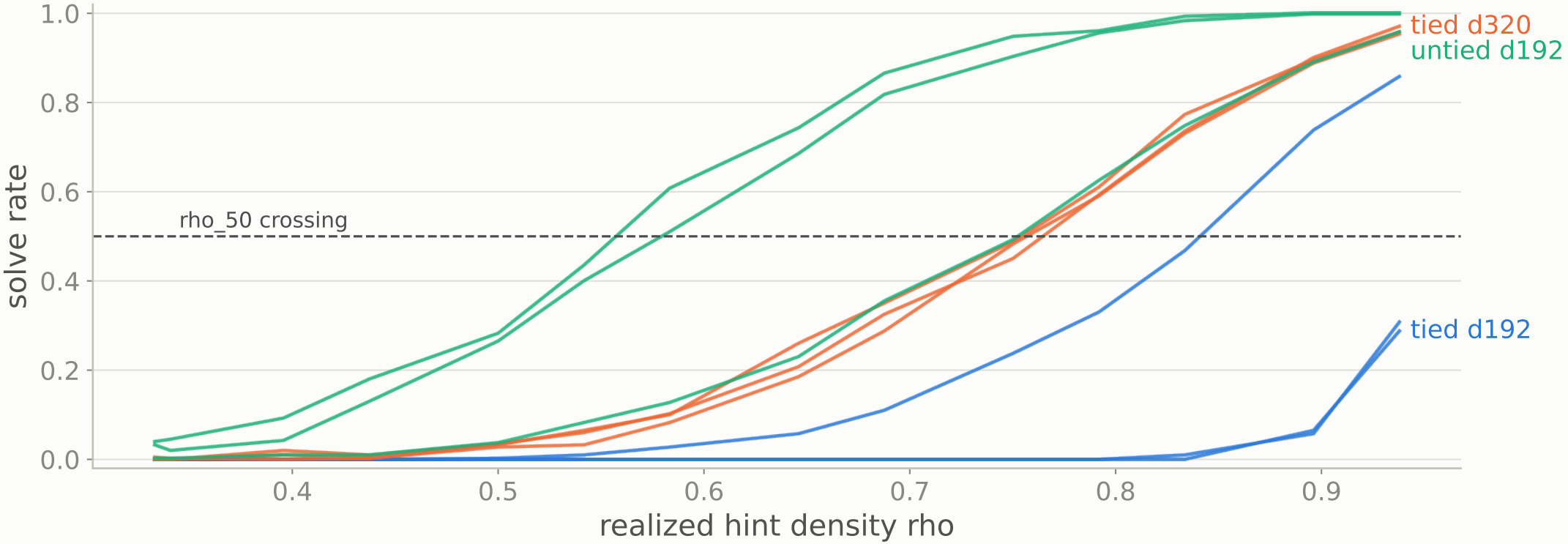
The model is never shown this reasoning -- only the node/colour pair.

A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the distribution every cell in this deck trained and was scored on.

T=5 rounds, the median of 40 draws from this config (range 1-19) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Tripling the width un-censors the tied cell: all three seeds cross 0.5



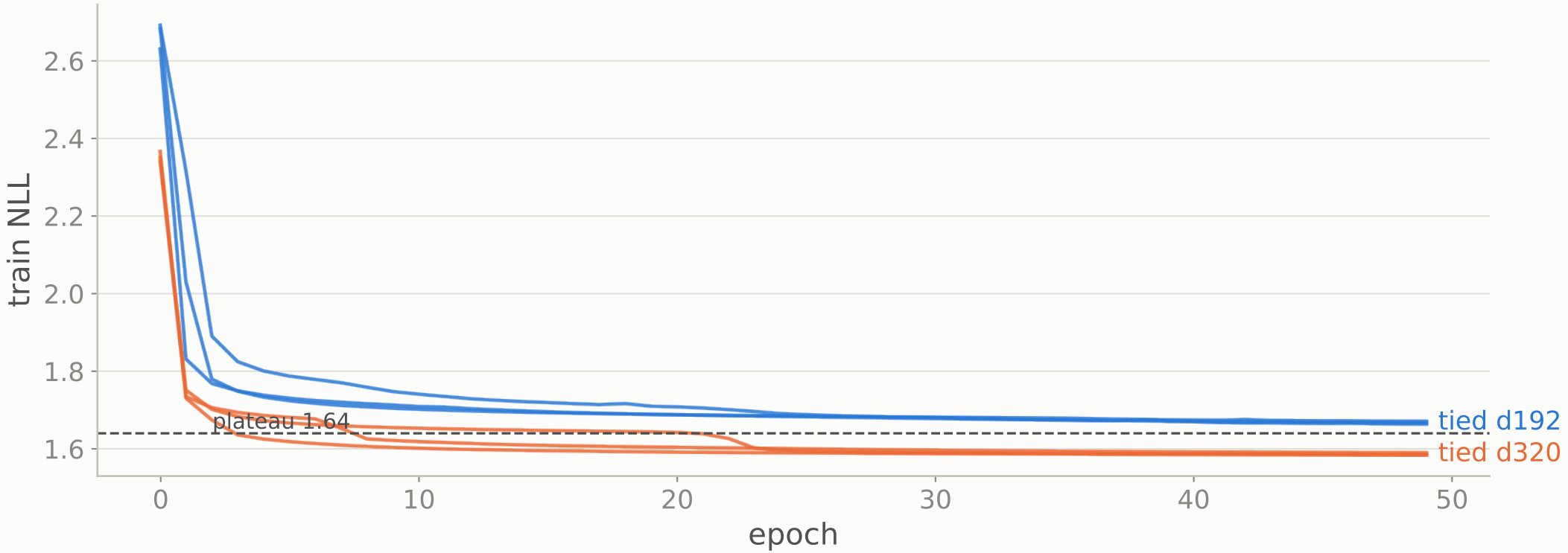
Three seeds drawn per cell, not a mean. Stage P's d192 never reaches the dashed line at any density.

The tied cell is 4.1x smaller than its untied twin and worse

cell	trainable	rho_50	mean solve	val acc	escapes
P1 tied d192	270,010	CENSORED	0.0268	0.4567	0/3
S1 tied d320	613,818	0.757	0.3148	0.4830	3/3
S2 tied d448	1,088,698	CENSORED	0.0000	0.0412	1/3
J2 untied d192	2,504,890	0.580	0.5527	0.4940	3/3

rho_50 LOWER is better. The untied twin J2 is the standing minimum on this eval; nothing here beats it.

The plateau was a capacity floor, not an obstruction from sharing



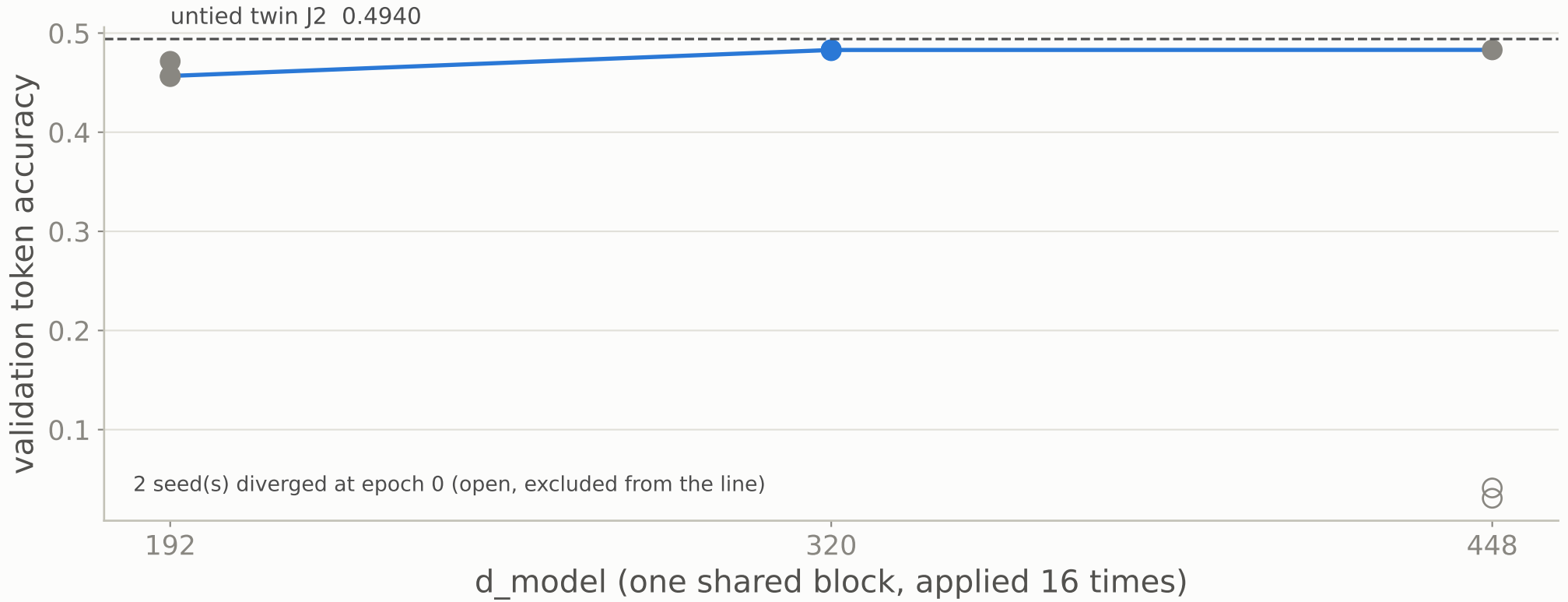
S1, read before any level: d192 sits on the plateau for all 50 epochs, d320 escapes it in every seed.

Every d320 seed escapes the plateau; no d192 seed does

cell	final train NLL by seed	escapes
P1 tied d192	1.6646 / 1.6721 / 1.6681	0/3
S1 tied d320	1.5851 / 1.5915 / 1.5864	3/3
S2 tied d448	1.5815 / 3.7874 / 3.9199	1/3

The plateau bar 1.64 was frozen in stage C and is reused here, not refrozen around this run.

Width buys accuracy up to d320, then stops short of the twin



d448's one trained seed is +0.0001 against d320's median; its other two never left epoch 0. Width has stopped buying accuracy.

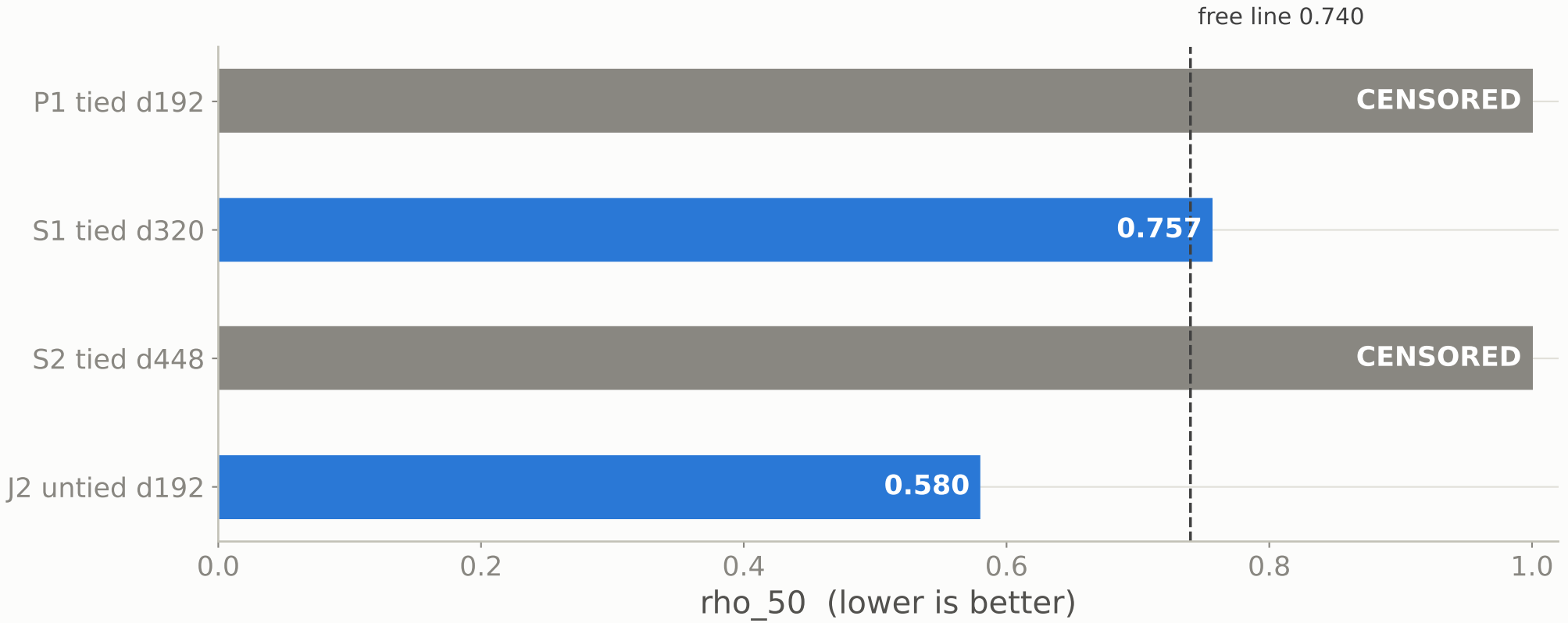
Two of three d448 seeds diverged at epoch 0, from warmup

- Diagnostic 51860479 traced it to warmup, not to d448 itself
- So d448 carries one seed and cannot support a 3-seed reading
- d320 is quoted throughout because d320 trained 3 of 3

The scorer printed S0 FALSIFIED; its own precedence rule outranks it

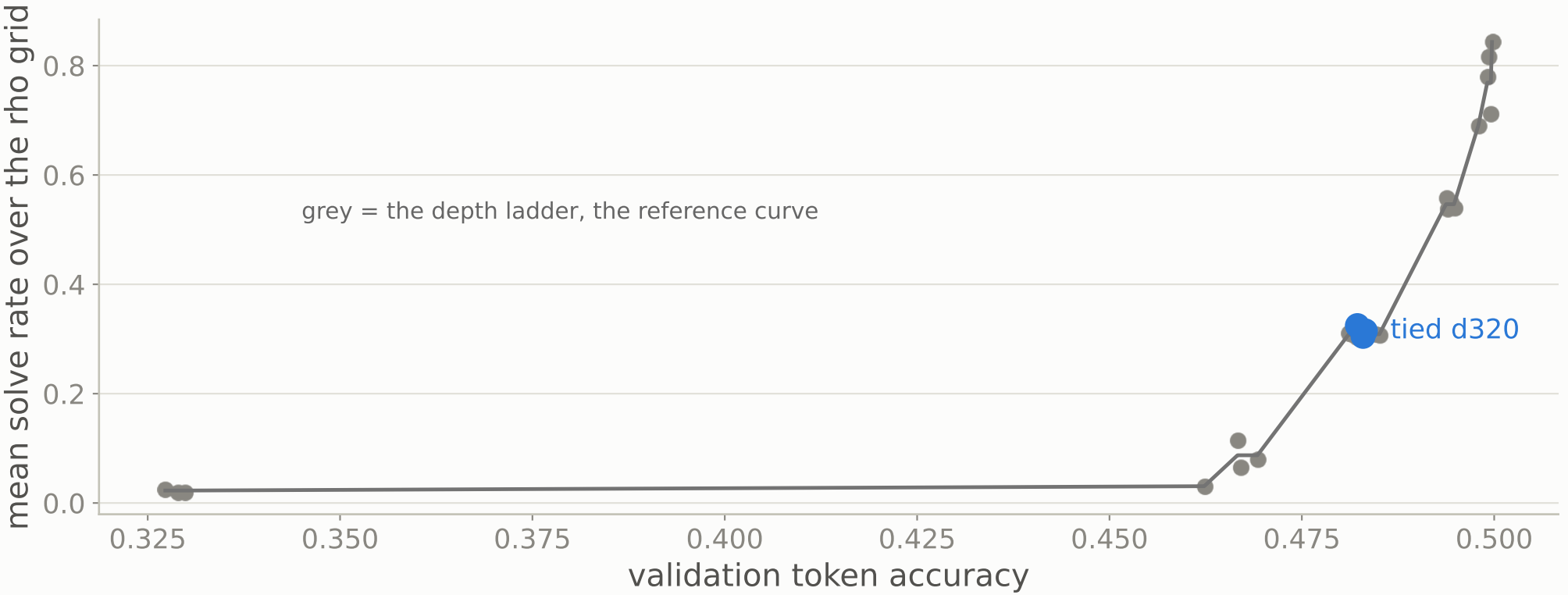
- S0 was registered on d448, which never trained: escapes 1/3, NLL 3.79
- S1 outranks S0: a cell that never left the plateau says nothing about tying
- d320's reading stands: 3/3 crossing, rho_50 0.757, 0.017 past the line

The tied cell lands 0.017 past the free line, not on it



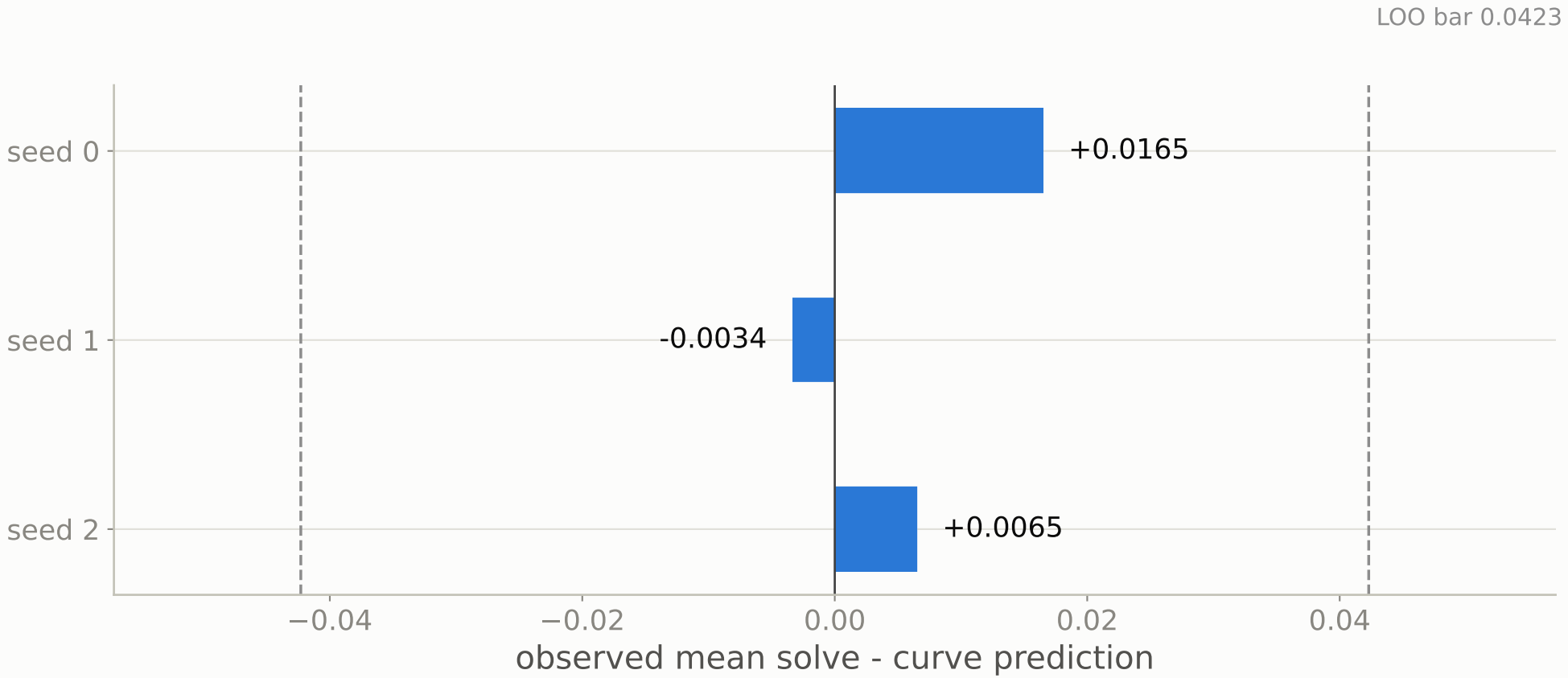
The free line 0.740 is where the unconstrained architecture solves. A level, not a bar this stage was registered to clear.

Tying moves the cell along the accuracy-to-solve curve, not off it



S4, registered before the run. The reference curve is the depth ladder's own isotonic fit; the tied cells were never in it.

All three tied seeds sit inside the bar that defines on-curve



Bar 0.0423 is stage J's leave-one-out residual, frozen there. Of 81 held-out cells 46 discriminate; the worst misses by 0.0729.

The tied cells are DISCRIMINATING, so the fit could have rejected them

seed	val acc	predicted solve	observed	residual	bracket
s0	0.4822	0.3079	0.3245	+0.0165	0.0008
s1	0.4830	0.3082	0.3048	-0.0034	0.0008
s2	0.4833	0.3083	0.3148	+0.0065	0.0008

A cell predicted near the 0.10 floor cannot fail this test; these are not. Bracket is the span between the two reference points.

The whole solve gap is what 0.011 of token accuracy buys

— Tied d320 0.4830 acc -> 0.757 rho_50; untied J2 0.4940 -> 0.580

— Both sit on one curve, so the gap is a LEVEL, not a relation

— Width does not close it: accuracy saturates 0.4830 -> 0.4837

Stage T: the tied block is a 16-step schedule, not an operator

- 16 is the count it was trained with, not a structural property
- Frozen weights, T=4..24: rho_50 bottoms at 16, then rises to 0.819
- T1 falsified: one seed's easy-bin solve_rate is 0.0000 at T=24

Stage S speaks only for N=48 at train depth 16

- Only N=48: the N=96 bars are a different size and appear nowhere
- Only T=16 at train time; no tied cell was trained at another depth
- No counterfactual audit of the tied cell; Q/R ran on untied

Against the previous best on this same eval, tying loses 0.177 rho_50

quantity	J2 untied d192 (standing)	S1 tied d320
rho_50 median	**0.580**	**0.757**
mean solve	0.5527	0.3148
val token accuracy	0.4940	0.4830
trainable	2,504,890	613,818
distinct blocks	16	1
seeds crossing 0.5	3/3	3/3

Same 14-point grid, same 400 instances per bin, same --shuffle-adj --fall. No improvement is claimed: the tied cell is simpler and worse.